

1. Introduction

With the rapid growth of digital education and online submissions, plagiarism has become a significant concern in academic environments. Students often submit assignments that may unintentionally or deliberately contain copied content from peers, books, or internet sources. Traditional manual checking methods are time-consuming, inefficient, and prone to human error, especially when handling a large number of submissions. Therefore, there is a need for an automated system that can efficiently analyze and detect similarities between documents.

The main objective of this system is to compare multiple student assignments, detect both direct and partial plagiarism, and generate accurate similarity reports. The system also classifies plagiarism levels using visual indicators and provides a user-friendly interface for uploading and analyzing assignments. Additionally, it highlights matched content to help users understand the similarity between documents clearly.

2. Literature Survey

Sr. No.	Author(s) & Year	Technique Used	Description of Work
1.	M. Potthast et al. (2014)	Text Alignment, NLP	Proposed the PAN plagiarism detection framework for identifying similarities between documents using text alignment techniques.
2.	A. Barrón-Cedeño (2013)	Text Similarity Analysis	Developed methods for detecting plagiarism using similarity measures and document comparison techniques.
3.	J. Devlin et al. (2019)	Deep Learning (BERT)	Introduced transformer-based models for understanding semantic similarity between texts.
4.	S. Ganguly et al. (2018)	TF-IDF, Cosine Similarity	Used vector space model to represent text and calculate similarity between documents.
5.	F. Grman & M. Ravas (2011)	Semantic Analysis (NLP)	Focused on detecting semantic similarity in documents using NLP techniques.

3. Problem Definition

In academic institutions, evaluating the originality of student assignments has become increasingly challenging due to the widespread availability of digital content. Students may copy material from online sources or from each other, either intentionally or unintentionally, leading to plagiarism.

Existing plagiarism detection methods are often manual or rely on basic keyword matching techniques, which are inefficient and unable to accurately detect paraphrased or partially modified content. These approaches are time-consuming, prone to human error, and difficult to scale when handling a large number of submissions.

Therefore, there is a need to develop an automated and intelligent system that can efficiently compare multiple documents, identify both direct and indirect plagiarism, and generate accurate similarity reports. The system should also be capable of highlighting matched content and providing a clear indication of plagiarism levels to assist educators in evaluation.

4. Objectives

The main objectives of this project are:

- a. To develop a web-based system for plagiarism detection .
- b. To compare multiple student assignments efficiently .
- c. To calculate similarity scores between documents .
- d. To classify plagiarism levels using color indicators .
- e. To provide a user-friendly interface for uploading and analyzing assignments .
- f. To highlight matching content for better understanding.

5. Proposed Work

The proposed system, Online Assignment Plagiarism Checker using Machine Learning, is developed as a web-based application that automates the process of detecting plagiarism in student assignments. The system is designed using a client-server architecture, where users interact through a web interface and the backend performs processing and analysis.

The system allows users to upload multiple assignment files in different formats such as PDF, DOCX, and TXT. These files are processed by the backend, where textual content is extracted using file handling libraries. The extracted text is then cleaned and normalized through preprocessing techniques such as lowercasing and removal of unnecessary spaces.

After preprocessing, the system applies Machine Learning-based techniques for text analysis. It uses TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual data into numerical feature vectors, which represent the importance of words in each document. These vectors are then compared using cosine similarity, which calculates the similarity score between documents.

To enhance detection accuracy, the system also incorporates:

- Character N-gram analysis to identify partially modified or paraphrased content
- Sentence-level comparison to detect matching lines between documents

The system compares each student's assignment with:

- Other student submissions
- Reference documents (if available)

Based on the comparison, a final similarity score is generated. The system then classifies the plagiarism level into categories such as:

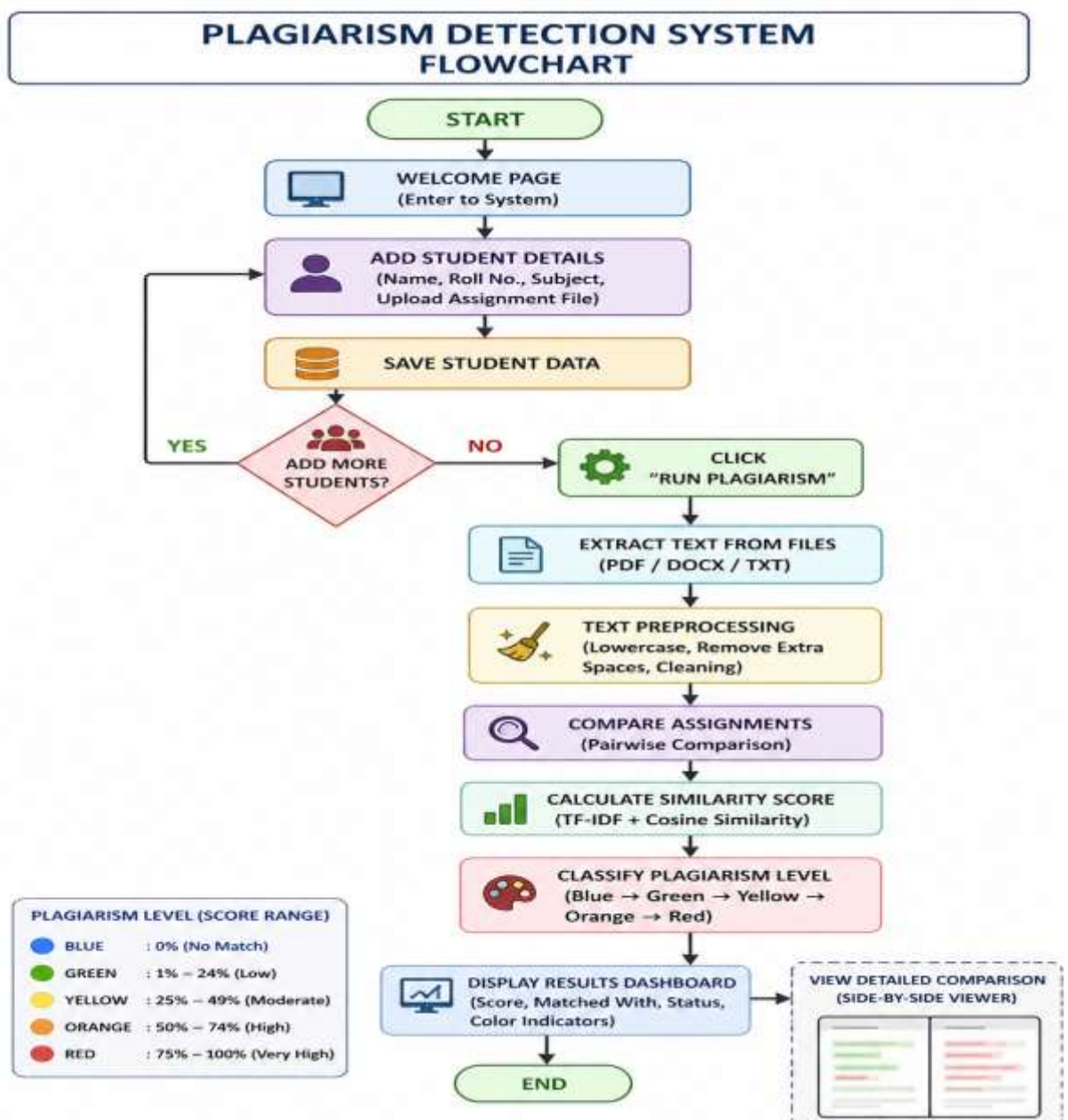
- High plagiarism
- Suspicious
- Original

The results are displayed through an interactive dashboard, which includes:

- Similarity percentage
- Visual indicators (progress bars / status labels)
- Highlighted matched content

This proposed system is designed to be efficient, user-friendly, and scalable for classroom-level use, enabling educators to quickly evaluate multiple assignments and maintain academic integrity.

Flowchart of the system:-



6. Methodology

6.1 Research Design

This project follows a design and development-based research approach. The system is designed as a web application using a client-server architecture.

The workflow includes:

1. Data input (assignment upload)
2. Text extraction from files
3. Preprocessing of text (cleaning and normalization)
4. Similarity computation using algorithms
5. Result visualization

The design ensures simplicity, efficiency, and usability

6.2 Data Flow Diagram

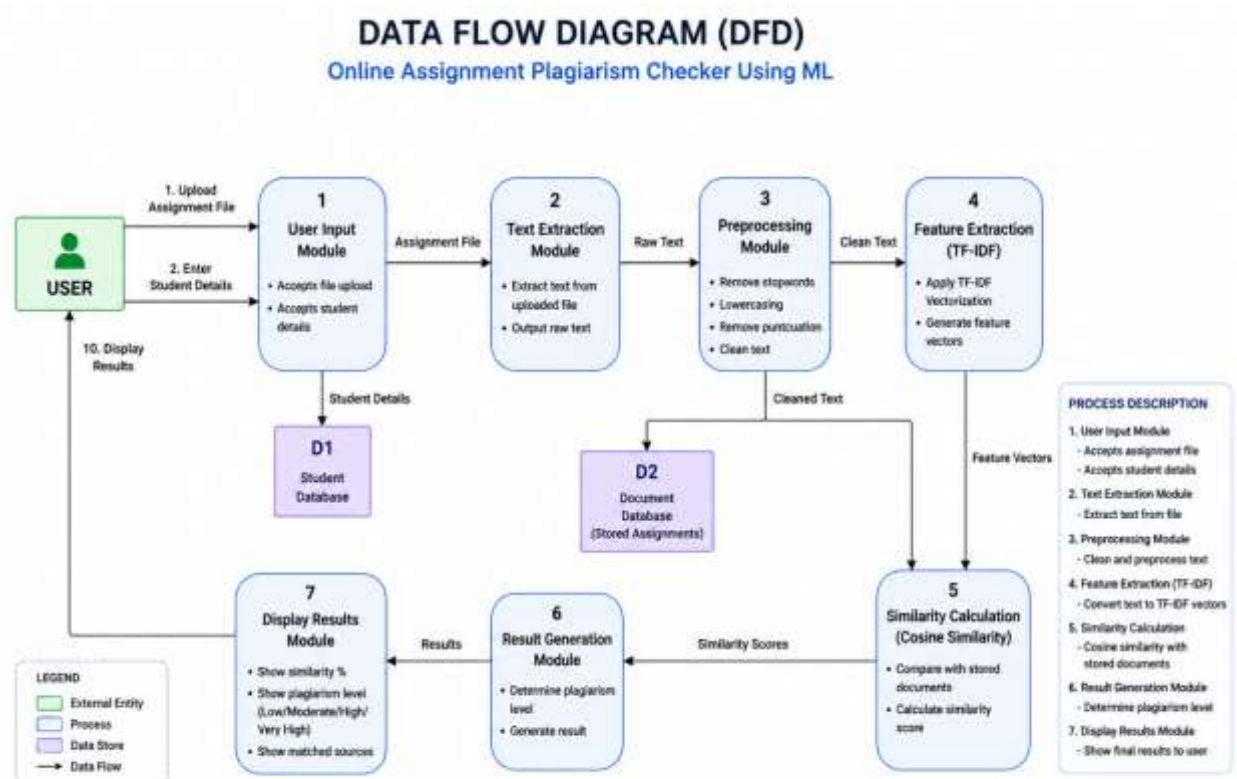


Fig.1 Data Flow Table

6.3 Data Collection Methods

The data used in this project consists of both user-uploaded assignments and a publicly available dataset obtained from Kaggle.

- **Kaggle Dataset:**

The system utilizes a plagiarism detection dataset available on Kaggle, which contains multiple text documents with varying levels of similarity. This dataset is used to understand and validate the effectiveness of Machine Learning techniques for plagiarism detection.

Dataset Link:

<https://www.kaggle.com/>

- **Student Assignments:**

In addition to the dataset, the system also processes real-time student assignment files uploaded by users in formats such as PDF, DOCX, and TXT.

The combination of a structured dataset and real-time input enhances the reliability and performance of the system.

6.4 Tools and Techniques Used

Tools Used:-

1. Python → Backend development
2. Flask → Web framework
3. HTML/CSS → Frontend design
4. VS Code → Development environment

6.5 Techniques Used

Text Preprocessing

1. Lowercasing
2. Removing extra spaces

TF-IDF Vectorization

1. Converts text into numerical form
2. Captures importance of words

Cosine Similarity

1. Measures similarity between two documents
2. Produces a percentage score

Character N-gram Analysis

1. Helps detect paraphrased content
2. Improves content-level similarity detection

7. Applications

This can be applied in various domains where content originality and document comparison are important. The major applications are as follows:

1. Academic Institutions

- Used in schools, colleges, and universities to check student assignments
- Helps teachers detect copied or similar submissions
- Reduces manual effort and improves evaluation efficiency

2. Assignment Evaluation Systems

- Automates the process of checking multiple assignments
- Compares student submissions quickly and accurately
- Ensures fair grading and transparency

3. Research and Thesis Verification

- Helps in verifying originality of research papers and theses
- Detects similarities with existing academic content
- Supports maintaining academic integrity

4. Online Learning Platforms

- Can be integrated into e-learning systems
- Automatically checks assignments submitted online
- Prevents academic dishonesty in virtual education

5. Content Writing and Publishing

- Used by writers and bloggers to check content originality
- Helps avoid duplicate or copied content
- Ensures uniqueness before publishing

6. Document Verification Systems

- Used in organizations to detect duplicate or reused documents
- Helps verify reports and official documents
- Useful for internal auditing and validation

7. Research and Development

- Acts as a base model for advanced plagiarism detection systems
- Can be extended using AI and deep learning techniques
- Useful for developing intelligent text analysis tools

8. Advantages

- **Automated Detection:-**The system eliminates the need for manual checking of assignments. It saves time and effort for educators and provides quick and efficient results.
- **High Accuracy:-**The system uses techniques such as TF-IDF and cosine similarity to ensure reliable comparison. It can detect both exact matches and partial similarities effectively.
- **Multi-File Support:-**The system supports multiple file formats including PDF, DOCX, and TXT. This makes it flexible and convenient for users to upload assignments in different formats.
- **Efficient Processing:-**The system can handle multiple assignments at the same time. It performs similarity calculations quickly, making it suitable for classroom use.
- **Detection of Partial Plagiarism:-**The system identifies not only direct copying but also modified or paraphrased content using character n-grams and sentence-level comparison.
- **User-Friendly Interface:-**The system provides a simple and easy-to-use interface. Users can upload assignments and view results without requiring technical knowledge.
- **Visual Result Representation:-**The results are displayed using similarity percentages, progress bars, and status indicators. This makes it easy to understand the level of plagiarism.
- **Improves Academic Integrity:-**The system helps maintain originality in assignments. It discourages plagiarism and promotes fair evaluation among students.

9. Limitations

- **Limited Semantic Understanding:-**The system mainly relies on text similarity and may not accurately detect meaning-based or conceptual plagiarism. It struggles when content is heavily paraphrased.
- **Sensitivity to Common Content:-**Common definitions, technical terms, or standard answers used by multiple students may increase similarity scores even when actual plagiarism is low.
- **Dependency on Text-Based Comparison:-**The system works only on textual data and cannot detect plagiarism in images, diagrams, or non-textual content.
- **No Permanent Data Storage:-**The system does not store data permanently, and results are temporary. This limits long-term tracking and comparison of assignments.
- **Scalability Issues:-**As the number of documents increases, the system may become slower due to pairwise comparison of assignments.
- **Limited Intelligence (No Deep Learning):-**The system uses basic machine learning techniques and does not incorporate advanced models like deep learning or transformers, which could improve accuracy.
- **Language Dependency:-**The system performs best with standard English text and may not work effectively for multiple languages or mixed-language content.

10.Future Scope

- **Integration with Advanced AI Models:-**The system can be enhanced by using advanced Natural Language Processing models such as deep learning and transformer-based models. This will improve the detection of semantic and meaning-based plagiarism.
- **Internet-Based Plagiarism Detection:-**The system can be extended to compare assignments with online sources, research papers, and websites. This will increase the coverage and accuracy of plagiarism detection.
- **Real-Time System Integration:-**The system can be integrated with online submission platforms or Learning Management Systems (LMS) to automatically check assignments at the time of submission and provide instant results.
- **Improved Semantic Analysis:-**Future improvements can focus on detecting idea-level similarity and conceptual plagiarism using advanced text analysis techniques such as topic modeling and keyword extraction.
- **Database Integration:-**A database can be added to store student assignments and maintain a history of submissions. This will allow long-term comparison and tracking of plagiarism patterns.
- **Enhanced Reporting System:-**The system can generate detailed and downloadable reports in formats such as PDF or Excel. Reports can include highlighted sections, similarity breakdown, and summary insights.
- **Multilingual Support:-**The system can be expanded to detect plagiarism in multiple languages, making it useful in diverse academic environments.

11.Expected Outcome

The proposed Online Assignment Plagiarism Checker using Machine Learning is expected to successfully detect similarities between student assignments and generate accurate plagiarism reports. The system will allow users to upload assignments, process the data, and display results through an interactive dashboard.

1.Login & Data Input

1. Login / Entry Page



This is the entry page of the system. User clicks on "ENTER SYSTEM" button to go to the dashboard.

2. Student Input Page



Here we add student details like name, roll number, subject and upload assignment file. Total number of students is displayed at the bottom.

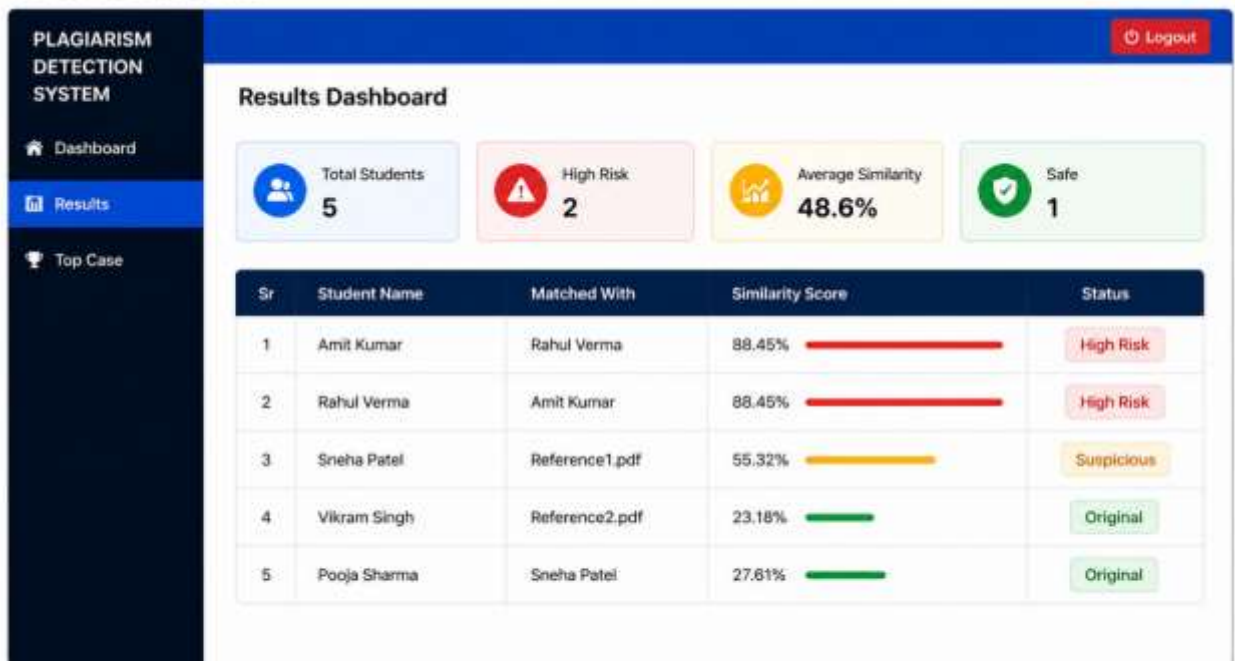
Fig.11.1: Login and Student Input Interface of the System

The system is expected to provide a user-friendly interface where users can enter student details and upload assignment files, as shown in Fig. 1. This interface simplifies the process of data input and ensures smooth interaction with the system. Furthermore, the system is expected to generate a results dashboard displaying similarity scores, matched documents, and plagiarism levels, as illustrated in Fig. 2. The results will be categorized using visual indicators such as:

- High Risk (Red)
- Suspicious (Yellow)
- Original (Green)

2.Result Dashboard:-

3. Results Dashboard



This is the main results dashboard. It shows all students with the file they are matched with, similarity percentage and their status. Red means high risk, yellow means suspicious and green means original.

Fig. 11.2: Results Dashboard Showing Similarity Scores and Status

The system will highlight matched content and display similarity percentages, helping users clearly understand plagiarism levels. Overall, the system is expected to be efficient, accurate, and capable of handling multiple assignments, thereby reducing manual effort and improving academic evaluation.

12. References

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